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Magnetic field effects dynamics of ethylammonium nitrate ionic liquid confined between glass plates

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Self-diffusion and NMR relaxation of ethylammonium (EA) cation were studied in the protic ionic liquid ethylammonium nitrate (EAN) confined between polar glass plates separated by a few μm distance and exposed to the external magnetic field of 9.4 T. The diffusion coefficient of EA (D) and the transverse NMR relaxation rate ($1/T_2$) of $-\text{NH}_3$ protons were increased immediately after placing the sample in the magnetic field by factors of ~ 2 and ~ 22 , respectively, in comparison with those of the bulk EAN. Further exposure of the sample to the magnetic field led to gradual changes of D , T_1 and T_2 to their bulk values with a time constant of ~ 70 min. Complete “recovery” of the sample to the “accelerated” D and “shortened” T_2 values occurred at longer than 24 hours after removing of the EAN sample from the magnet. Because the observed characteristic times of the change far exceed the times of molecular processes in EAN, we suggested that this phenomenon is related with reversible phase transformations occurring in confined EAN.

Keywords: *Ionic liquid; Nuclear magnetic resonance; Pulsed-field gradient, Self-diffusion; NMR relaxation; Ion dynamics*

Introduction

Ionic liquids (ILs) are molten salts prepared from organic cations and either organic or inorganic anions.^{1,2} Their applications are continuously expanding, including uses, for instance, as electrolyte material in lithium batteries³ and ultracapacitors,⁴ as media for chemical reactions and separation,^{2,5} and as lubricants.⁶ Ethylammonium nitrate (EAN) is a protic ionic liquid. This means that it has an available proton on the cation.¹ EAN first synthesized by Paul Walden in 1914,⁷ is the most frequently reported protic IL.¹ It is used as a medium for chemical reactions, as a precipitating agent for protein separation,⁵ an electrically conductive solvent in electrochemistry,³ and other applications. Like water, EAN has a three-dimensional hydrogen-bonding network and can be used as an amphiphilic self-assembly medium.⁸

Ions of ILs can participate in a variety of interactions that may lead to formation of ordered self-assembled structures.⁹ External conditions, such as temperature, pressure, electric field and confinement, may change the structural and dynamic parameters of ILs.^{9,10-13} Confinement in vertically aligned carbon nanotubes,¹⁴ mesoporous carbon¹⁰ and silicon^{11,12} does increase the diffusion of ILs by a factor of 2-3 or more. The latter has been explained as a consequence of decreasing the packing density of ions in pores with diameters comparable to the sizes of the ions.¹¹ Recently, enhanced diffusion of ethylammonium (EA) cations has been observed for EAN confined between polar glass plates separated by 4 μm ,¹⁵ which has been related to the transformation of a sponge-like structure proper to bulk EAN. Magnetic field strength is also an important external variable that influences the properties of some organic substances, such as liquid crystals¹⁶ and organic-based semiconductors.¹⁷ No perceptible magnetic field effects have been observed for non-magnetic ionic liquids until now.^{9,18}

In this work, we report the effect of applying a strong magnetic field to EAN confined between polar glass plates separated by $\sim 4\text{ }\mu\text{m}$. This system demonstrated accelerated translational diffusion of EA cations and accelerated transverse NMR relaxation rates of EA $-\text{NH}_3$ protons.¹⁵ After exposing this system to a strong magnetic field (9.4 T), we observed a slow (with a time constant of ca 70 min) decrease in the diffusion coefficient of the EA cation and a gradual increase in T_2 NMR relaxation times for EA $-\text{NH}_3$ protons toward bulk values of EAN. No such effect was observed either.

Experimental

Sample preparation

EAN (see Figure 1) consists of an EA cation and a nitrate anion. EAN used in our experiment was synthesized, pretreated and characterized at the Chemistry of Interfaces of Luleå University of Technology as described previously.¹⁵

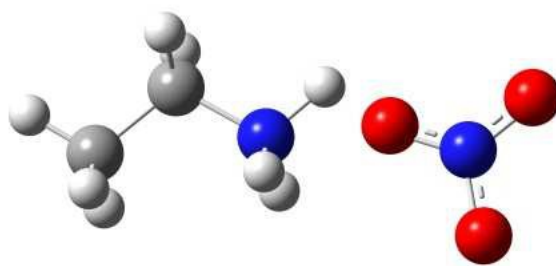


Figure 1. Chemical structure of EAN, consisting of an ethylammonium (EA) cation (left) and a nitrate anion (right).

NMR measurements for bulk EAN were performed by placing it in a standard 5-mm NMR glass tube. Confined EAN was prepared with thin glass plates arranged in a stack (see Figure S1 in the Electronic Supporting Information (ESI)). The plates (5 x 14 x

0.1 mm, Thermo Scientific Menzel-Gläser, Menzel GmbH, Germany) chemical composition presented in ESI. They were carefully cleaned before use (see ESI). Contact angle measurements with Milli-Q water provided a contact angle near 0° (see ESI). A stack of glass plates filled with EAN was prepared in a glove box in a dry nitrogen atmosphere. Samples were prepared by adding 2 μL of EAN to a first glass plate, placing a second glass plate on top, adding 2 μL to this glass plate, etc. until the thickness of the stack reached approximately 5 mm. Excess EAN from the sides of the stack was removed by wiping. The sample consisted of a stack of ca 45 glass plates with EAN between them, placed in a rectangular sealed glass tube. The mean spacing between glass plates was estimated by weighing the introduced EAN, which yielded $d \sim 3.8\text{--}4.5 \mu\text{m}$.¹⁵ A detailed description of the sample preparation and characterization has been reported in our previous paper.¹⁵

EAN with deuterium substituted $-\text{NH}_3$ protons was prepared by mixing of the original EAN with D_2O followed by water evaporation under high vacuum ($\sim 10^{-3}$ mBar) during 48 h. Degree of substitution was determined from the intensity of the corresponding line in the ^1H NMR spectrum.

Pulsed-field gradient diffusometry

^1H NMR self-diffusion, T_1 and T_2 relaxation measurements of EAN, in bulk and confined between glass plates, were performed using a Bruker Ascend/Aeon WB 400 (Bruker BioSpin AG, Fällanden, Switzerland) NMR spectrometer with a working frequency of 400.27 MHz for ^1H , magnetic field strength of 9.4 T with magnetic field homogeneity better than $4.9 \cdot 10^{-7}$ T, using a Diff50 Pulsed-Field-Gradient (PFG) probe. A NMR solenoid ^1H insert was used to macroscopically align the plates of the sample stack at 0 and 90 degrees with respect to the direction of the external magnetic field.

The diffusional decays (DD) were recorded using the spin-echo ($t_d \leq 5$ ms) or the stimulated echo ($t_d > 5$ ms) pulse sequences. For single-component the signal intensity, A , changes as:^{19,20}

$$A(\tau, g, \delta) \propto \exp\left(-\frac{2\tau}{T_2}\right) \exp(-\gamma^2 \delta^2 g^2 D t_d) \quad (1a)$$

$$A(\tau, \tau_1, g, \delta) \propto \exp\left(-\frac{2\tau}{T_2} - \frac{\tau}{T_1}\right) \exp(-\gamma^2 \delta^2 g^2 D t_d) \quad (1b)$$

for the spin-echo and stimulated echo, respectively. Here τ and τ_1 are time intervals in the pulse sequence; γ is the gyromagnetic ratio for protons; g and δ are the amplitude and the duration of the gradient pulse; $t_d = (\Delta - \delta/3)$ is the diffusion time; $\Delta = (\tau + \tau_1)$. In the measurements duration of 90° pulse was $7\mu\text{s}$, δ was in the range 0.5-2 ms, τ was in the range 3-5 ms, and the amplitude of g was varied from 0.06 up to $29.73 \text{ T}\cdot\text{m}^{-1}$. The recycle delay was 3.5 s.

As has been shown in our previous work,¹⁵ there is no effect of restrictions on D in the direction along the plates (see Figures S2 and S3 in the ESI), while for diffusion normal to the plates the effect of the plates on D is noticeable at t_d longer than 3 ms (see Figures S4, S5 and S6 in the ESI). For diffusion in bulk and along the plates, D values were obtained by fitting Eqs. (1) to the experimental decays. For decays obtained in the direction normal to the plates at $t_d = 3$ ms, D was calculated from the equation:

$$D_{av} = \frac{-\partial A(\gamma^2 \delta^2 g^2 t_d)}{\partial (\gamma^2 \delta^2 g^2 t_d)} \Big|_{(\gamma^2 \delta^2 g^2 t_d) \rightarrow 0} \quad (2)$$

All measurements were performed at 293 K and were started immediately (in 1-2 minutes) after placing the sample in the NMR probe. The time required for a single

NMR diffusion measurement was around 30 s, while that of a relaxation measurement was around 3 min. Data were processed using Bruker Topspin 3.5 software.

To demonstrate the absence of effect of the background gradient g_0 in such heterogeneous system as glass plates with the confined ionic liquid, a 13-interval stimulated echo sequence with bipolar gradient pulses²¹ (SteBp2) was used. The latter pulse sequence removes cross-terms arising from the applied pulsed gradients, g and g_0 , which are significant even if g_0 values are small.

¹H NMR relaxometry

¹H T_1 and T_2 NMR relaxation time measurements were performed with inversion-recovery (180° - τ - 90° -fid) and CPMG (90° - τ - 180° - τ -echo) pulse sequences, respectively.²² Duration of 90° pulse was 7 μ s. Phases of 90° and 180° pulses as well as the phase of receiver in the CPMG pulse sequence were varied with a specific cycle, and echo time was fixed to allow elimination of diffusion and J -coupling modulation effects.

Results and discussion

Since all the measurements are performed at room temperature (293 K), the measurements were performed immediately after placing the sample in the magnet. Placement of bulk EAN in the magnetic field of the NMR spectrometer and systematic measurements of D , T_1 and T_2 over 24 hours showed no changes to these parameters (Table S1 in ESI). However, for EAN confined between glass plates, we observed changes in the D , and T_1 and T_2 of different chemical groups of EA, as shown in Figures 2.

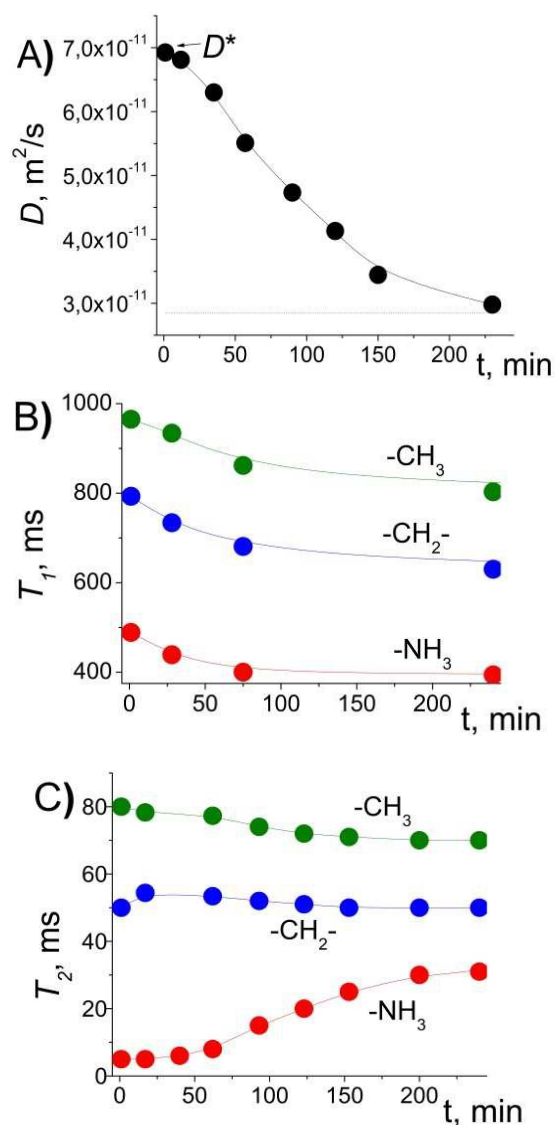


Figure 2. Change of A) Diffusion coefficient of EA cations, B) T_1 NMR relaxation times and C) T_2 NMR relaxation times of protons of different chemical groups of the EA cation after placing EAN confined between polar plates in a magnetic field of 9.4 T. D was measured along the plates, $t_d = 3$ ms.

Right after placement the sample in the NMR probe, D was measured as D^* , which was a factor of ~ 2 larger than that in bulk (D_0). T_1 relaxation recovery for protons of different groups demonstrated forms close to the exponential ones. T_2 relaxation for the -NH₃ group was exponential, while T_2 relaxation of -CH₂- and -CH₃ groups were close to the sum of the two exponential functions (see Figure S7 in the ESI). In this case, the

mean values of T_2 were calculated as: $T_2 = (1/T_{21} + 1/T_{22})^{-1}$. Initial values of T_2 (T_2^* , right after placement of the sample in the magnetic field) of different groups are decreased by factors of ~ 8.8 ($-\text{CH}_2-$), ~ 12.1 ($-\text{CH}_3$) and ~ 22 ($-\text{NH}_3$) relative to their bulk values. Further, we observed that D gradually decreased from D^* to lesser values close to D_0 (Figure 2A). The same effect was observed either by applying unipolar pulse sequences (spin echo and stimulated echo) or by applying stimulated echo sequence with bipolar gradient pulses (SteBp2), Figures S8 and S9 in the ESI. Therefore, the effect is not related with background gradient g_0 of magnetic field. This agrees with the NMR spectra linewidths, which should be broadened in the presence of the background gradient. In fact, the linewidths typical for bulk maintains for EAN between glass plates.¹⁵

After placing the sample in a magnetic field, simultaneously with D , the T_1 of different protons decreased by a factor of ~ 1.25 (Figure 2B). Mean values of T_2 for $-\text{CH}_2-$ and $-\text{CH}_3$ groups demonstrated small changes, while $T_2(-\text{NH}_3)$ increased by a factor of ~ 10 (Figures 2C and S10 in the ESI). A similar effect was observed for the sample oriented along and normal to the glass plates relative to the magnetic field (Figure 3, black circles and stars).

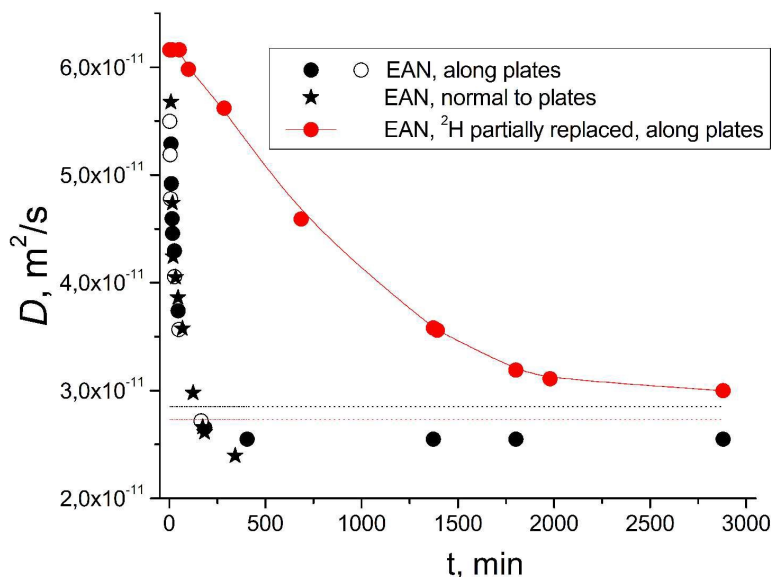


Figure 3. Change of diffusion coefficient of ionic liquids placed between polar glass plates: EA cations in original EAN (black symbols), diffusion of EA cation in EAN with ~30% protons of -NH_3 groups replaced with deuterium (red circles). Dotted lines show bulk values of D for corresponding ionic liquids.

From Figure 2A, one can see that the decrease in D and the increase in $T_2(\text{-NH}_3)$ plateau after ca 4 hours. The process can be characterized by the time during which the initial value changes by a factor of “ e ”, which was estimated from Figure 2A to be nearly 70 min. The experiment was repeated several times with different time intervals, TE, between completion of the change in D and T_2 values in the previous experiment and the beginning of the next experiment. In cases when TE was just few hours, the measured D value in the beginning of the new experiment was smaller than D^* , and $T_2(\text{-NH}_3)$ was larger than $T_2^*(\text{-NH}_3)$. But in cases when TE was on the order of a day or longer, D and T_2 were completely “recovered” to D^* and to T_2^* , respectively.

Observed acceleration of diffusion in the beginning of the experiment cannot be related with a “mechanical relaxation” of the sample in the strong magnetic field, which can lead to a flux of EAN. In that case, the measured (“apparent”) diffusion

coefficient would be related with molecular displacements over longer distances, therefore it would be higher than the true molecular diffusion coefficient. This would lead also to increase of the apparent D with the increase of t_d . Here, there is no dependence of diffusion decays (and measured diffusion coefficient) on the t_d (Figure S3, solid symbols). Therefore measurements were performing at equilibrium conditions.

Additional experiments were performed to ensure that the effects observed for D and $T_2(-\text{NH}_3)$ are due to the presence of a strong magnetic field, but not associated with other processes that may occur in the NMR spectrometer during diffusion and relaxation experiments. To investigate the effect of radio-frequency pulsing, a fully “recovered” sample was placed in the spectrometer probe for a rather long time, but no radio-frequency pulses were applied during this time interval. The first measurements of D and T_2 were performed after 24 hours and yielded the same values as those from experiments where continuous NMR measurements were taken. In another experiment, a sample that had “recovered” outside of the magnetic field of the spectrometer was placed between the poles of Neodymium permanent disk-shaped magnets with a magnetic field strength of ~ 0.68 T for 24 hours before placing the sample in the NMR spectrometer for measurements. After placing the sample in the spectrometer, the measured value of D was between D^* and D_0 . Therefore, one can conclude that the presence of a strong magnetic field is the main factor influencing the D and $T_2(-\text{NH}_3)$ processes that occur with EAN confined between polar plates.

To reveal main factors leading to the observed effect of alteration of diffusion coefficient in the presence of magnetic field few extra samples were prepared. For one of the samples we used silanized nonpolar glass plates. Silanization protocol is described in ESI. In this case D in the EAN layers between glass plates was close to D_0

and no alteration occurred after placement of the sample in magnetic field. For preparation of another sample we used nonprotic ionic liquid $[P_{6,6,6,14}][BOB]^{23}$ (Figure S11 in ESI) and polar glass plates. Diffusion of $[P_{6,6,6,14}]^+$ cation in layers between plates was just slightly lower than that in bulk and no any effect of magnetic field was observed. Therefore magnetic field effects only on diffusion of protic IL placed between polar plates.

State and dynamics of liquids between solid surfaces are attracting scientific interest for many years. Earlier studies of interaction between charged plates led to discovery of the Casimir effect.^{24,25} On a submicron scale, this force becomes the dominant force between uncharged conductors. Several recent experiments have reported data for the approach of two surfaces separated by ionic liquids which revealed relatively long range forces (tens of nm) that appear to be electrostatic in origin.²⁶ The original interpretation of these forces was controversial, and the origin of long range forces in ionic liquids remains the subject of discussion. No forces acting in the μm range, as well as no effects of magnetic field were discovered. Earlier we discovered acceleration of diffusion of EAN occurring between glass plates and suggested that this is related to exchange processes of mobile $-\text{NH}_3$ protons.¹⁵ To further explore this phenomenon, we substituted ca. 30 % of these protons by deuterium by mixing of EAN with D_2O followed by water evaporation under high vacuum. It is well known that exchange rates of deuterium in chemical exchange reactions are considerably slower than those for protons, phenomena usually called as the kinetic isotope effects. Indeed, for a sample containing EAN (with partially substituted protons by deuterons) confined between the glass plates, the rate of decrease of D after placement of the sample in a magnetic field of 9.4 T was more than a factor of 10 slower as compared to

non-deuterated EAN samples (Figure 3, red circles). Therefore, the exchange rate of -NH_3 protons plays a crucial role in the observed processes.

The time-scale of the magnetic field phenomenon observed in EAN confined between polar plates is on the order of a few hours, which is much longer than the time-scale of dynamic processes occurring with individual EAN ions confined between glass plates. Indeed, the longest of the dynamic processes is the self-diffusion of the ions, which covers the inter-planar distance during ca. 100 ms. For this reason, we suggested that the processes occurring with EAN between glass plates is the reversible phase transformations of EAN. It has been suggested that EAN, because of the competition of the strong hydrogen bonding and electrostatic interactions of EA cations and nitrate anions, and the hydrophobic interactions of $\text{-CH}_2\text{-}$ and -CH_3 groups of EA cations, may form a bi-continuous sponge-like structure.^{9,27-29} Because this structure is characteristically a balance of forces, structural parameters can vary as a consequence of changes in temperature, pressure and chemical modification of ions. Hindered diffusion is typical for this structure, as well as for the bi-continuous cubic phase, because of the curvilinear trajectories of ions along the micro-phase borders.³⁰ It could be suggested that the balance of forces and structural parameters of the sponge-like phase may be modified in the presence of additional interactions of ions with the polar surfaces of glass plates. The change in structural parameters enhances the mobility of the ions.^{14,31} Another possibility is the transformation of the sponge-like phase to an isotropic phase. The phase transformation leads to enhancement for diffusive transport and for accelerated -NH_3 proton exchange. Application of a strong magnetic field additionally contributes to inter-ion interactions, which might be more efficient for moving ions, leading to further disturbance of the existing equilibrium.

Thus, the application of a strong magnetic field to EAN confined between polar plates allows one to control the dynamics of EA ions.

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Conclusions

In this work, we have experimentally shown that self-diffusion and NMR relaxation of EAN confined between polar glass plates reversibly alters after application of strong magnetic field. Main factors conditioned this effect are availability of protons in the protic EAN and polarity of the surface. Exchange rate of -NH_3 protons plays a crucial role in the observed processes. Because of long time scale of this process, we suggested that the processes occurring with EAN between glass plates is the reversible phase transformations of EAN. Further structural studies are required to support this suggestion.

Given that the ionic liquid dynamic properties are significantly altered in the confinement and in the external magnetic field, these results have strong implications for interface-intensive applications of ILs, such as their use as lubricants and in electrochemical and electro-magnetic systems.

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